Alon: Audio Speech Evaluator and Recommendation System for Effective Communication

A Research Proposal

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Chapter 1 – BACKGROUND OF THE STUDY

Problems and Opportunities

Oral communication is a core competency in higher education and in the workplace. Public speaking is a significant element of many college courses and is strongly related to students' academic performance and future career success (Lintner & Belovecová, 2024). Still, many university students feel anxious when speaking in front of others, and it is one of the most common anxieties among them. The pressure is even greater during thesis defenses, class presentations, project pitches, and job interviews, where nervousness often gets in the way of a good performance.

Even so, the ways speakers currently get feedback still have several gaps.

1. Public speaking evaluation usually depends on subjective human ratings, takes a lot of time and effort, and is not always available during routine practice.
2. automated transcription tools can identify what is said, but transcription alone does not fully explain how the speech is delivered.
3. speakers may need more specific feedback that identifies where speech issues such as filler words, repeated words, repeated phrases, and unnecessary pauses occur during the recording. Providing these issues with timestamps can help users locate the parts of their speech that need more attention.

Recent studies also show the importance of assessing speech under different speaking conditions. Sun (2023) used both read aloud and spontaneous speaking activities and explained that spontaneous speech may reveal pronunciation and speaking difficulties that

may not appear during controlled reading. This is important because a speaker who performs well while reading prepared text may experience different difficulties when forming and delivering an answer independently. Established speaking assessments also use different types of speaking activities. The TOEIC Speaking test includes tasks such as reading a text aloud, describing a picture, responding to questions, responding using provided information, and expressing an opinion (Educational Testing Service, n.d.). These approaches provide a basis for Alon to use different speaking activities in its initial assessment instead of relying on only one type of speech task.

The initial assessment of Alon will consist of five speaking stages that increase in task complexity. The assessment will begin with more controlled speaking activities and progress toward tasks that require the user to form responses independently. The results will establish the user's baseline speech delivery before regular practice. This allows succeeding attempts to be compared with the initial performance and gives the user a clearer view of changes in speech delivery over time.

These gaps create an opportunity for innovation. Combining Natural Language Processing (NLP) with Audio Signal Processing allows the system to analyze linguistic features such as transcription, filler word, repeated word, and repeated phrase detection and acoustic features such as pause duration, jitter, shimmer, and speaking rate at the same time. The system can then generate a timestamped vocal delivery heatmap that shows the moments where speech issues occur. A recommendation component can also turn the detected speech issues into practical coaching feedback. In addition, accepting microphone input and pre-recorded audio in English makes the tool accessible for private, self-paced practice without a human coach.

The application is named Alon, the Filipino word for “wave.” The name reflects the nature of speech as sound that travels through waves and the way vocal patterns change throughout a recording. It also represents the progress of the user across repeated speaking attempts, where changes in speech delivery can be observed over time. Through its speech analysis, timestamped results, vocal delivery heatmap, and coaching feedback, Alon aims to help users recognize patterns in their delivery and identify areas that need further practice.

Statement of the Problems and Hypothesis

This research seeks to assess the performance of Alon, an automated system for speech analysis and coaching, in evaluating oral communication delivery and presentation metrics. The study specifically aims to address the following questions:

1. How accurately can Alon measure speaking rate, jitter, and shimmer from recorded speech to generate a timestamped vocal delivery heatmap?

2. How accurately can Alon transcribe English speech and detect filler words, repeated words, and repeated phrases in academic and professional speaking contexts?
3. How relevant and useful are Alon's automated recommendations in helping users identify and improve detected speech issues?
4. How acceptable is Alon to students and educators based on the following ISO/IEC 25010 quality characteristics?
    - 4.1 Functional Suitability
    - 4.2 Usability
    - 4.3 Performance Efficiency
    - 4.4 Reliability

Hypotheses

Beneficiaries

Students and Job Seekers. They can use Alon to review their speech delivery and identify areas for improvement before thesis defenses, class presentations, and job interviews. The application's transcript, vocal delivery heatmap, and automated recommendations can help them recognize frequent filler words, and repeated words and phrases, and changes in their vocal patterns.

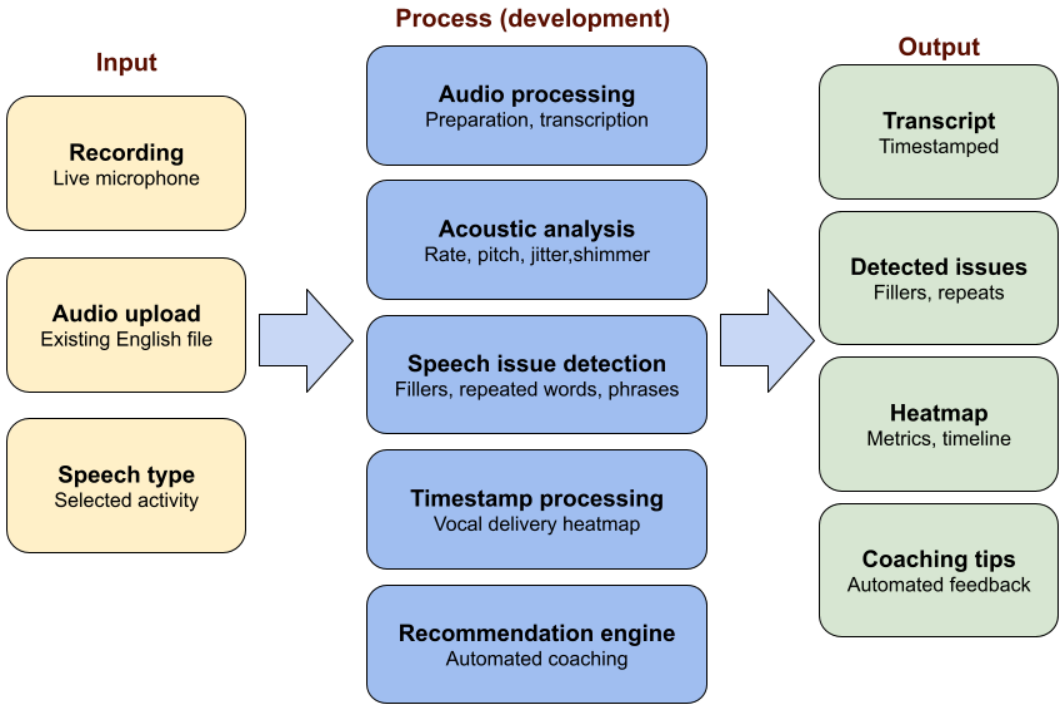
Sales and Marketing Professionals. They can use Alon to practice product presentations, sales pitches, and client communication. The application's speech analysis and feedback can

help them become more aware of their speaking pace, vocal variation, and frequent speech repetitions.

University Career Centers and Educators. They may use Alon as a supplementary tool during presentation practice, mock interviews, and communication training. Its speech analysis results and automated feedback can support existing assessment methods, although they should not replace professional judgment or established evaluation rubrics.

Researchers and Data Scientists. This study may serve as a reference for future research involving audio signal processing, speech transcription, vocal feature analysis, and automated recommendation systems. It may also provide insights into the potential use of speech analysis technologies in educational and professional development applications.

Chapter 2 – ANALYSIS OF RELATED LITERATURE AND STUDIES AND CONCEPTUAL FRAMEWORK



The conceptual framework of Alon follows the Input-Process-Output model. In the input stage, the user gives Alon a speech by either recording it with a microphone or uploading an existing English audio file, and chooses the type of speaking activity, such as an academic presentation, research defense, or mock job interview. In the process stage, Alon prepares the audio and turns it into a transcript, then measures the speaking rate, jitter, and shimmer. It also detects filler words, repeated words, and repeated phrases, attaches timestamps to the results so the user can find the exact parts of the recording, builds the vocal delivery heatmap, and generates automated coaching recommendations. In the output stage,

the user receives the timestamped transcript, the detected speech issues, the acoustic measurements and heatmap, and the coaching tips.

Review of Related Literature and Studies

A. **Public Speaking Anxiety and Vocal Delivery**

B. **Real-Time Speech and Anxiety Prediction**

C. **Acoustic Feature Analysis & Emotion Recognition**

D. **Lexical Features and Disfluencies**

E. **Fusion of Audio Signal Processing and Natural Language Processing (NLP)**

Synthesis

Researchers have emphasized the relevance of computational speech analysis in evaluating public speaking anxiety, recognizing emotional patterns, and assessing speaking effectiveness. In exploring the prediction of public speaking anxiety during oral presentations, it highlights the importance of analyzing anxiety-related patterns throughout a speech (Kimani et al., 2022). This widens the scope of how it can affect an individual's public speaking efficiency; thus, even speaking privately for specific situations can be affected by anxiety. With the demonstration of different acoustic features of an individual in speech such as pitch, jitter, and shimmer, in speech emotion recognition using a deep learning approach (Bhangale and Kothandaraman, 2023). In a similar context, Rozgic et al. (2012) looked at the integration of lexical and acoustic features and proposed that linguistic information and vocal traits can offer complementary data for speech analysis. Furthermore,

frequent speech disfluencies—especially filler sounds—can have a detrimental effect on how effective public speaking is perceived, according to Laske and DiGennaro Reed (2024). When taken as a whole, these studies offer a theoretical and technical basis for using automated techniques to analyze various facets of oral presentation performance. It can help the researchers study Alon to make data gathering more efficient, in which factors can be taken in consideration for the detection of inefficiencies in order to create an accurate and useful system.

Chapter 3 – METHODOLOGY

Scope of the Study

Research Design

The researchers will use both experimental research designs. In this part, the researchers will build and test Alon. The researchers will record audio, upload it, transcribe speech, pull out acoustic features, detect filler words, process timestamps, classify speech, create a vocal delivery heatmap, and generate feedback.

In this part, the researchers will see how Alon works with different speech recordings. The researchers will use recordings from students and benchmark speech samples that we find online.

Subject and Locale

The participants are students from the University of the East Manila. They will use Alon while doing practice presentations, research defense drills, or mock job interviews.

Sampling

The researchers will use purposive sampling for the student participants. The researchers will pick students who can give an English presentation, practice a research defense, or do a mock job interview.

The researchers will also use purposive sampling for benchmark recordings. Every benchmark recording must have English speech from a single main speaker and enough continuous speech for analysis. The researchers will skip recordings that are heavily distorted and have overlapping speakers, missing audio, or background music that drowns out the speaker. The researchers will prefer recordings that have pacing, few filler words, few unnecessary pauses, and smooth speech delivery.

Data Gathering Techniques

The researchers will collect data from student recordings, benchmark recordings, and Alon evaluations. Each recording will give us numbers for speech rate, pause duration, pitch variation, jitter, shimmer, filler words, verbal crutches, and timestamps.

The researchers will check speech rate by using a human-verified transcript and the real duration of the chosen speech segment. The researchers will count the words and calculate reference words per minute. Then, the researchers will compare that number with the speech rate from Alon.

For pause duration, the researchers will look at selected recordings and mark when each pause starts and ends. Those marks will be our reference pause measurements. The researchers will then compare Alon's detected pause durations with those reference values.

A human-verified transcript will be our reference for filler words and verbal crutches. The researchers will label the occurrences in chosen speech samples. Then, the researchers will compare Alon's detections with those labels using precision, recall, and F1 score.

The researchers will also compare Alon's generated transcript with a human-verified transcript using Word Error Rate.

Paper Prototype

It displays the user interfaces and reports designs for each system user. The presentation of the paper prototype should be chronologically arranged based on the flow of operations.

Data Visualization

It provides a discussion of the raw data that will be collected and/or data fields from the proposed prototype that can be used to generate the required outputs for visualization and interpretation.

Bibliography

This section should identify the different sources of information utilized in the accomplishment of the project. All sources mentioned in the document should be reflected here. Sources are categorized by type (books, journals, theses, etc.). and should be written alphabetically according to the Author's Surname. Use the appropriate APA format for proper bibliographical entries.

Appendices

This section allows the group to include the documents related to their topic. This may include sample documents, prototypes, interviews, survey instruments (if applicable), etc.

SAMPLE APA FORMAT FOR TABLE

Table 1

Rating, Range, and Its Verbal Interpretation

Rating	Range	Verbal Interpretation
5	4.51 – 5.00	Highly Acceptable
4	3.51 – 4.50	Acceptable
3	2.51 – 3.50	Moderately Acceptable
2	1.51 – 2.50	Slightly Acceptable
1	1.00 – 1.50	Not Acceptable

Note: Table description here

FORMAT:

- **Paper Size :** Letter
- **Font :** Arial, 12 pts, Justify Alignment
- **Line Spacing :** Double
- **Margin :** Left – 1.5”, Top, Right, Bottom – 1”
- **Page Number :** Top Right, Arial 9 pts

Similarity Index : **20%**

AI Report : 20%

H_0: Alon does not meet the predefined performance and acceptability criteria for speech analysis, speech issue detection, timestamped vocal delivery visualization, feedback generation, and selected ISO/IEC 25010 quality characteristics.

H_1: Alon meets the predefined performance and acceptability criteria for speech analysis, speech issue detection, timestamped vocal delivery visualization, feedback generation, and selected ISO/IEC 25010 quality characteristics.

Check sub tab/s.

1. REVIEW OF RELATED LITERATURE

2. Sun, W. (2023). The impact of automatic speech recognition technology on second language pronunciation and speaking skills of EFL learners: A mixed methods investigation. Frontiers in Psychology, 14, 1210187. https://doi.org/10.3389/fpsyg.2023.1210187
3.

4. REVIEW OF RELATED STUDIES

2.1 Real-Time Public Speaking Anxiety Prediction

Kimani et al. (2022) presented a study titled *Real-time Public Speaking Anxiety Prediction Model for Oral Presentations*. The researchers investigated computational prediction of public speaking anxiety during oral presentations. Their work is directly related to examining anxiety-related patterns during a speech rather than evaluating the entire recording as a single unit.

The study supports the feasibility of developing systems that analyze public speaking anxiety over time. Alon builds on this direction by proposing a timestamped vocal anxiety heatmap, which would allow users to locate specific segments for further practice. The study is relevant to the system's general objective, although Alon's specific combination of jitter, shimmer, speaking rate, NLP, and automated recommendation must be tested independently.

2.2 Speech Emotion Recognition Using Multiple Acoustic Features

Bhangale and Kothandaraman (2023) developed a speech emotion recognition approach that combines multiple acoustic features with a one-dimensional deep convolutional neural network. Their feature set included pitch, jitter, shimmer, Mel-frequency cepstral coefficients, and other signal characteristics.

Their work demonstrates how several measurable properties of a speech signal can be combined in a machine learning model. This is relevant to Alon's audio pipeline, which proposes extracting vocal characteristics to identify segments that may warrant closer review. However, emotion classification performance does not automatically establish anxiety-detection accuracy, making task-specific validation important.

2.3 Filler Words and Perceived Speaking Effectiveness

Laske and DiGennaro Reed (2024) conducted an experimental study examining how filler sounds and words affect perceived public speaking effectiveness. They prepared speeches with varying rates of disfluency and had participants evaluate the speaker's performance. The results showed that higher disfluency rates, especially frequent filler sounds, negatively affected several measures of perceived effectiveness.

This study provides empirical support for including filler-word detection in Alon. The system could count fillers, identify their locations in the transcript, and indicate whether users repeatedly rely on particular expressions. Rather than treating every filler as an error, Alon could provide contextual feedback based on frequency and speaking patterns.

2.4 Acoustic and Lexical Feature Fusion for Emotion Recognition

Rozgic et al. (2012) examined a speech emotion recognition method that combined acoustic information with lexical features derived from speech recognition output. The researchers used multiple feature representations and evaluated their effectiveness for recognizing emotional categories.

The study is relevant to Alon's proposed integration of audio signal processing and NLP. It suggests that acoustic characteristics and linguistic content can provide complementary information. Alon could similarly combine measurements such as pitch variation and speaking rate with transcription results and filler-word counts to produce a more comprehensive speech evaluation. However, the model's usefulness would depend on how well these combined features predict the particular delivery problems that Alon is designed to identify.

REFERENCES

Bhangale, K., & Kothandaraman, M. (2023). Speech emotion recognition based on multiple acoustic features and deep convolutional neural network. Electronics, 12(4), Article 839. https://doi.org/10.3390/electronics12040839

Kimani, E., Bickmore, T., Picard, R. W., Goodwin, M., & Jimison, H. (2022). Real-time public speaking anxiety prediction model for oral presentations. In *Companion Publication of the 2022 International Conference on Multimodal Interaction* (pp.

160–164). Association for Computing Machinery. https://doi.org/10.1145/3536220.3563686

Laske, M. M., & DiGennaro Reed, F. D. (2024). Um, so, like, do speech disfluencies matter? A parametric evaluation of filler sounds and words. *Journal of Applied Behavior Analysis, 57*(3), 574–583. https://doi.org/10.1002/jaba.1093

Rozgić, V., Ananthakrishnan, S., Saleem, S., Kumar, R., Vembu, A. N., & Prasad, R. (2012). Emotion recognition using acoustic and lexical features. In *Proceedings of Interspeech 2012* (pp. 366–369). https://doi.org/10.21437/Interspeech.2012-118

Speech contains various acoustic characteristics that can provide information about a speaker's emotional and physiological state. Among the commonly analyzed acoustic features are fundamental frequency, jitter, shimmer, intensity, and other low-level descriptors. These characteristics can change depending on a person's emotional condition, making speech analysis a potential non-invasive approach for identifying stress and anxiety.

Pallewela et al. (2024) investigated the use of advanced audio cues for speech emotion recognition through machine learning. Their study examined acoustic characteristics such as fundamental frequency, jitter, and shimmer and demonstrated their usefulness in identifying emotional intensity in speech. The findings suggest that measurable variations in vocal characteristics can provide meaningful information regarding a speaker's emotional state. This is relevant to the present study because jitter and shimmer can potentially serve as objective acoustic indicators when analyzing speech associated with anxiety or stress.

Vocal tremors and changes in vocal stability may also occur when an individual experiences heightened emotional pressure. During situations such as public speaking, anxiety may affect voice production, resulting in noticeable changes in pitch, intensity, and vocal steadiness. These observations provide a basis for incorporating acoustic analysis into systems designed to identify possible anxiety-related speech patterns.

Furthermore, Murali et al. (2022) developed a real-time public speaking anxiety prediction model that utilized audio features to predict moments of anxiety during oral presentations. Their research is particularly relevant because it demonstrates the feasibility of analyzing speech over time rather than treating an entire presentation as a single observation. Such an approach supports the concept of identifying specific periods in a presentation when anxiety indicators become more prominent.

REFERENCES:

Pallewela, N., Alahakoon, D., Adikari, A., Pierce, J. E., & Rose, M. L. (2024). Optimizing speech emotion recognition with machine learning based advanced audio cue analysis. Technologies, 12(7), 111. https://doi.org/10.3390/technologies12070111

Problems and Opportunities

Lintner, T., & Belovecová, B. (2024). Demographic predictors of public speaking anxiety among university students. *Current Psychology*. https://doi.org/10.1007/s12144-024-06216-w

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SCOPE OF THE STUDY

This study covers the development and evaluation of Alon: Audio Speech Evaluator and Recommendation System for Effective Communication, a speech coaching application designed to help students and other individuals practice oral presentations, research defenses, and mock job interviews.

The application will allow users to record their speech through a microphone or upload an existing audio file for analysis. Alon will examine selected speech characteristics, including speaking rate, jitter, and shimmer. It will also identify selected filler words, repeated words, and repeated phrases in English speech. These features are intended to help users identify patterns in their speech delivery that may require improvement.

After processing the audio, Alon will generate a speech transcript and associate the analysis results with timestamps so users can locate specific portions of the recording for review. The application will also present a timestamped vocal delivery heatmap that visualizes changes in selected acoustic characteristics throughout the recording. Based on the detected speech patterns, Alon will provide automated coaching recommendations to help users improve their speaking pace, recognize frequent speech repetitions, and become more aware of their vocal delivery.

Alon's primary user is the speaker. Users can record or upload speech, submit recordings for analysis, review generated transcripts, examine detected speech issues, view acoustic measurements and the vocal delivery heatmap, and read the application's recommendations.

Delimitations

The study will only focus on English-language speech. Filipino, Taglish, and other languages will be excluded from the intended language coverage. The application’s transcription and analysis performance will depend on the capabilities of the selected speech recognition and audio processing technologies.

Alon will focus exclusively on speech analysis. It will not analyze facial expressions, posture, eye contact, gestures, body movements, or other visual behaviors. It will also not assess the factual correctness, logical structure, persuasiveness, or academic validity of a presentation’s content. Furthermore, the application will not diagnose anxiety, stress disorders, or other medical or psychological conditions. The selected acoustic measurements will describe observable patterns in the recording and will not independently establish a speaker’s emotional or psychological state.

The evaluation of Alon will focus on the accuracy of its speech analysis features, the relevance of its coaching recommendations, and its acceptability in terms of functional suitability, usability, performance efficiency, and reliability, as guided by the applicable ISO/IEC 25010 software quality model.

RESEARCH DESIGN

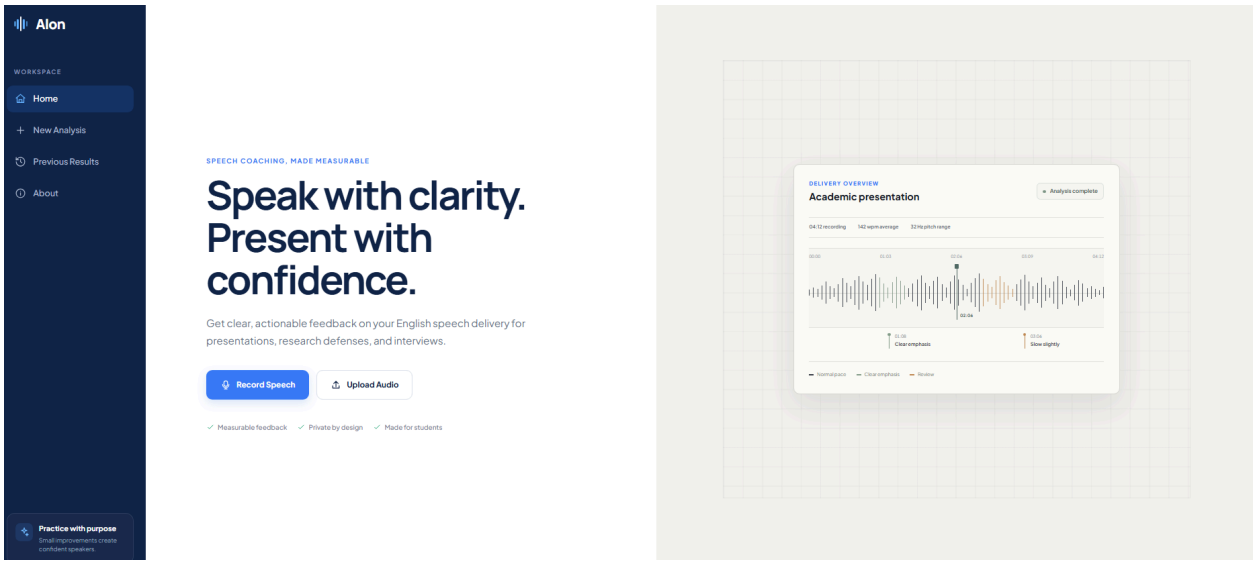
The researchers will use a developmental research approach combined with quantitative testing and evaluation to develop and assess Alon: Audio Speech Evaluator and Recommendation System for Effective Communication. The development phase will involve building the application and integrating its audio processing, speech transcription, speech issue detection, timestamp processing, vocal delivery heatmap, and automated recommendation features.

The testing phase will assess Alon's performance using recordings from student participants and suitable English-language benchmark speech samples. The researchers will compare the system's outputs with manually verified transcripts and reference measurements obtained using appropriate audio-analysis tools. The evaluation will also examine the relevance and usefulness of Alon's recommendations and its acceptability to users in terms of functional suitability, usability, performance efficiency, and reliability, guided by ISO/IEC 25010.

SUBJECT AND LOCALE

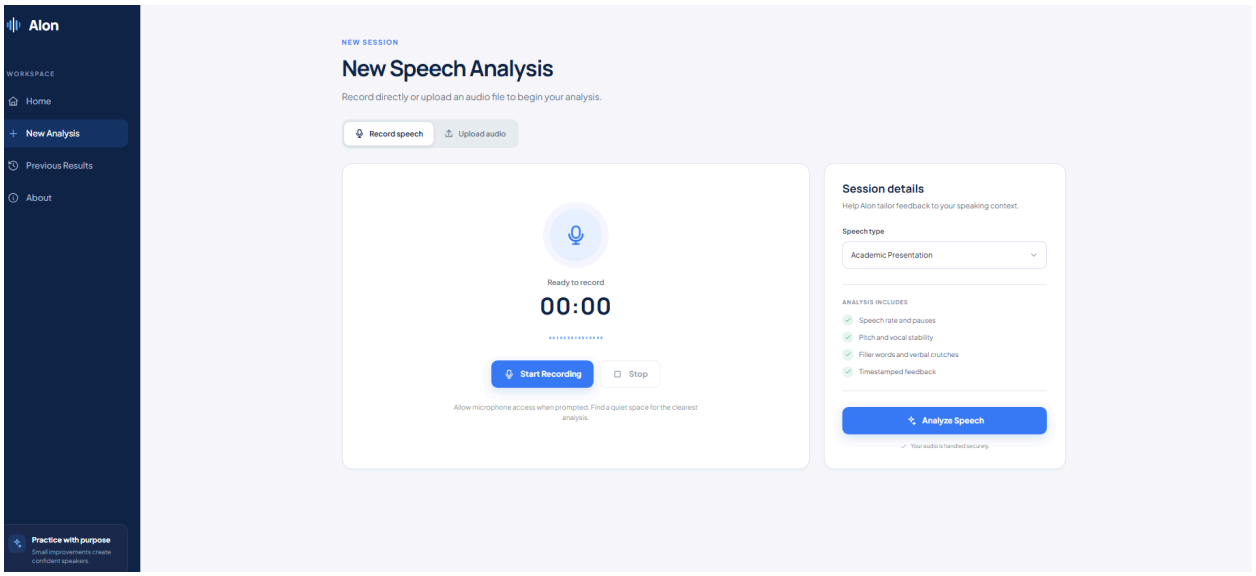
The participants of this study are first-year students of the University of the East, Manila. The study will involve two groups: first-year Information Technology students and first-year Broadcasting students. The first-year Broadcasting students will serve as the control group, and the first-year Information Technology students will serve as the comparison group. The two groups will be compared to examine differences in their speech delivery results as measured by Alon.

Figure 1. Alon Home Screen



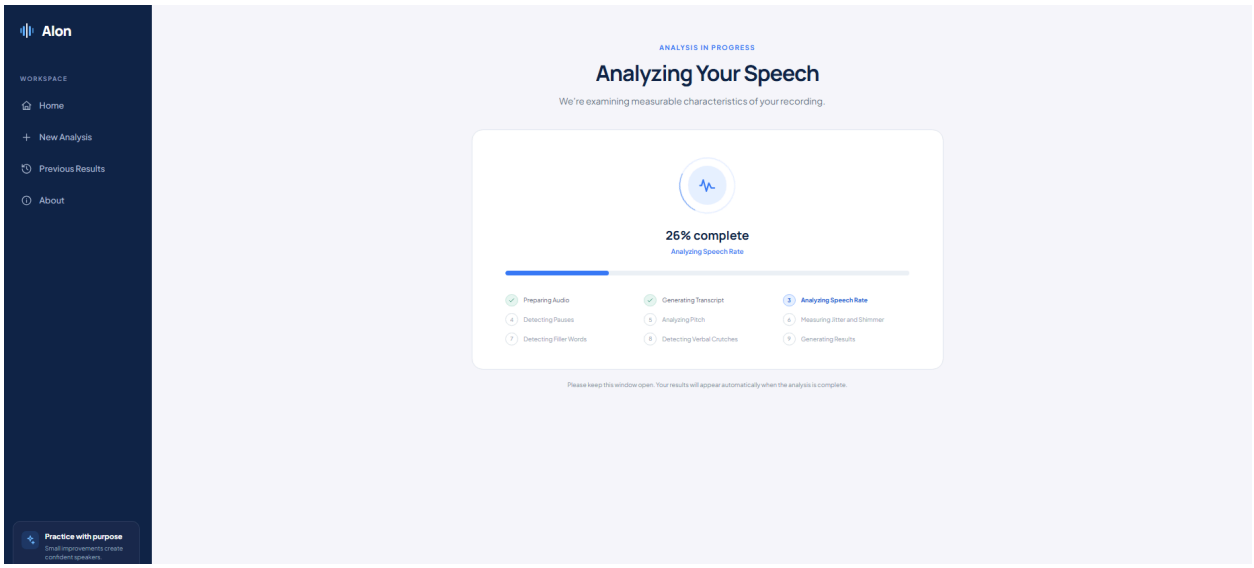
The home screen serves as the starting point of Alon. It introduces the application and allows the user to begin a speech analysis by recording a new speech or uploading an existing audio file. The user can also access previous results and information about the application.

Figure 2. New Speech Analysis Screen



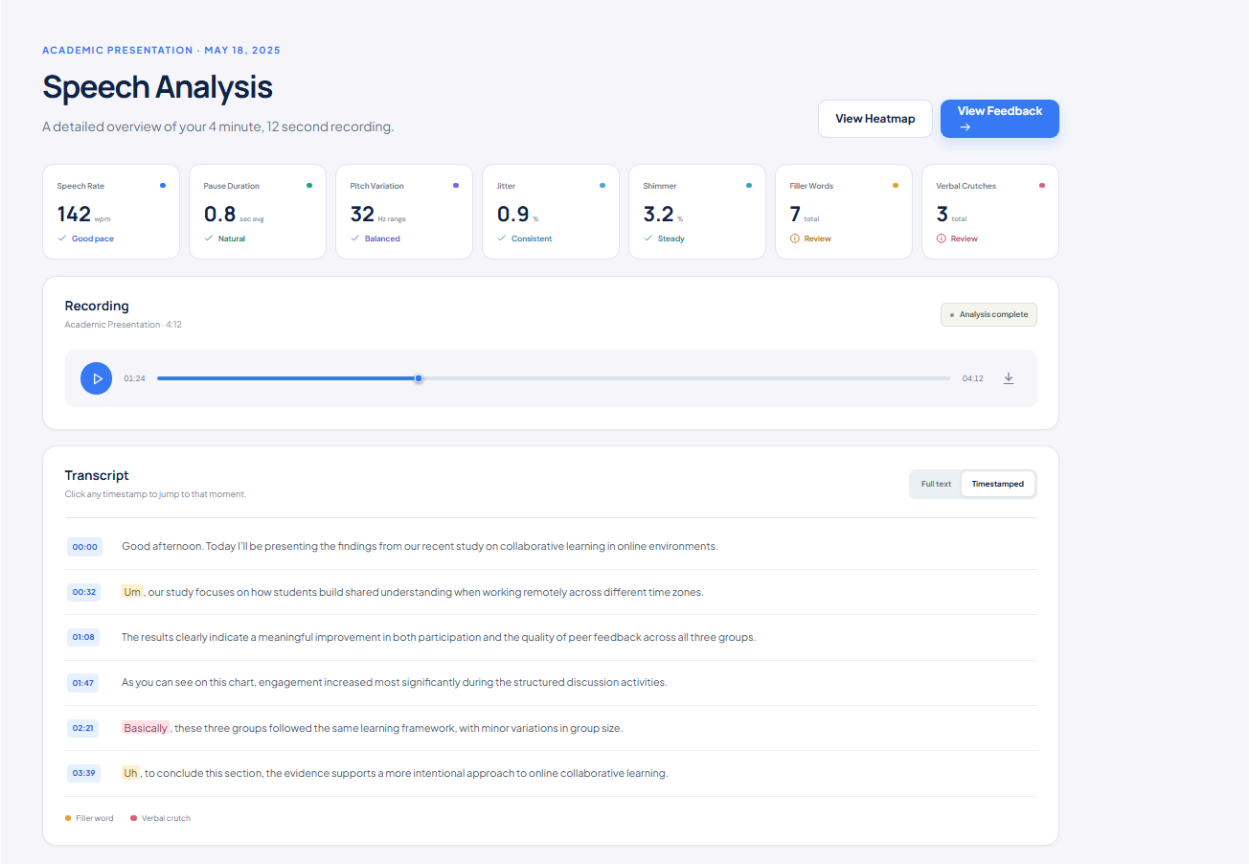
The New Speech Analysis screen allows the user to record speech using a microphone or upload an existing audio file. The user can also select the type of speaking activity, such as an academic presentation, research defense, or mock job interview, before submitting the recording for analysis.

Figure 3. Speech Processing Screen



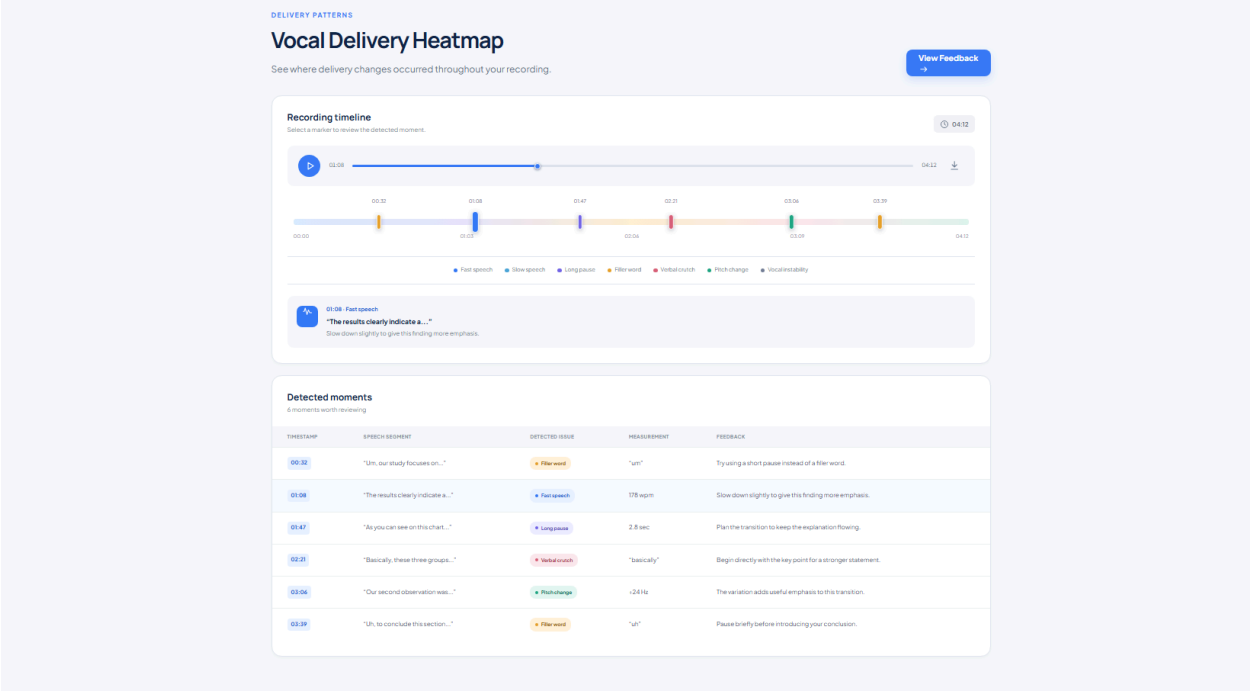
The speech processing screen informs the user that Alon is analyzing the submitted recording. During this stage, the application prepares the audio, generates the transcript, calculates the speech rate, detects pauses, measures jitter and shimmer, detects filler words and verbal crutches, and prepares the results.

Figure 4. Speech Analysis Results Screen



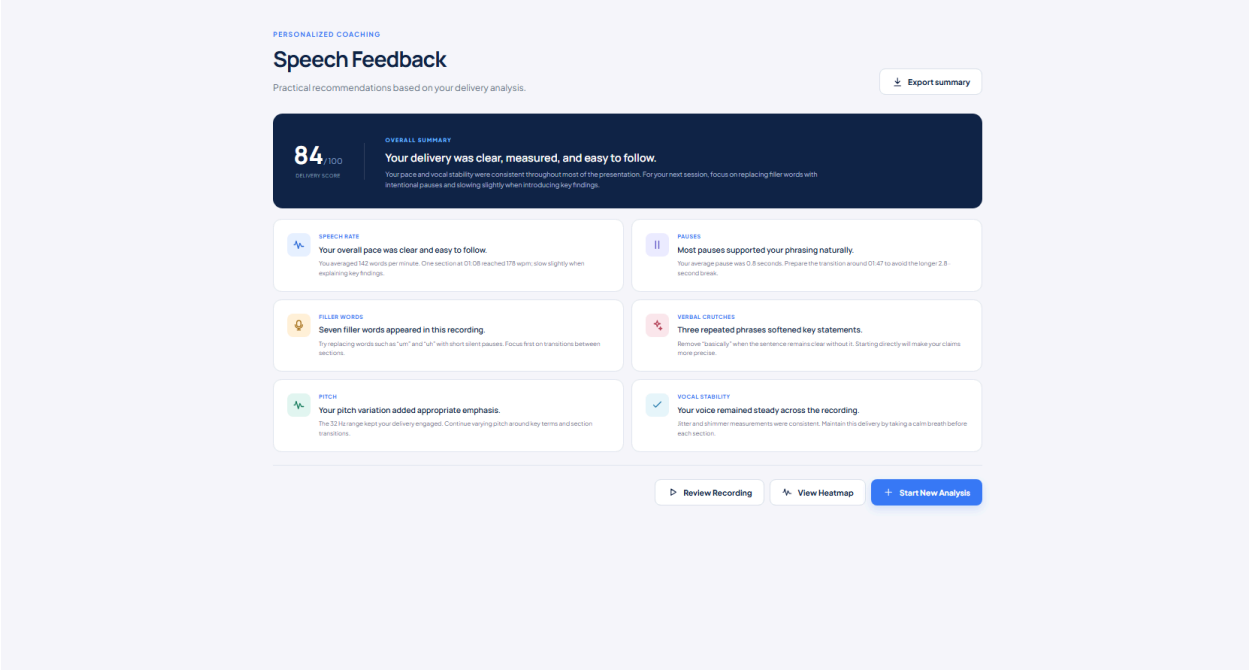
The Speech Analysis Results screen presents the measurements generated from the user's recording. It displays speech rate, pause duration, jitter, shimmer, filler words, and verbal crutches. The screen also provides an audio player and a timestamped transcript that allows the user to review where detected filler words and verbal crutches occurred.

Figure 5. Vocal Delivery Heatmap Screen



The Vocal Delivery Heatmap screen presents detected speech delivery issues according to their position in the recording. The timeline allows the user to review issues such as changes in speech rate, long pauses, filler words, and verbal crutches. Detected moments are also shown with their timestamps, speech segments, measurements, and corresponding feedback.

Figure 6. Speech Feedback Screen



The Speech Feedback screen provides recommendations based on the speech analysis results. Feedback is presented for speech rate, pauses, filler words, verbal crutches, and vocal stability based on jitter and shimmer measurements. The user can review the recording, return to the heatmap, or begin another speech analysis.

GOOGLE FORMS 1: PRELIMINARY NEEDS ASSESSMENT

Purpose: Gather information about students' public-speaking experiences and the features they would find useful in Alon.

Alon: Student Public Speaking Experience and Needs Assessment

This questionnaire aims to understand students' experiences, challenges, and needs related to public speaking. The results will help inform the development of Alon, a proposed speech-analysis and coaching system. Participation is voluntary(?), and responses will be handled confidentially.

Section 1 — Respondent Profile

1. What is your current year level?
    - 1st year
    - 2nd year
    - 3rd year
    - 4th year
    - Other
2. How often do you participate in public-speaking activities?
    - Very often
    - Often
    - Sometimes
    - Rarely
    - Never
3. Which activities have you participated in? (Select all that apply)
    - ☐ Class presentations
    - ☐ Oral reports
    - ☐ Debates
    - ☐ Speech competitions
    - ☐ Job interviews
    - ☐ Other

Section 2 — Public Speaking Challenges

Response scale:
1 — Strongly disagree
2 — Disagree
3 — Neither agree nor disagree
4 — Agree
5 — Strongly agree

I have difficulty speaking confidently in front of an audience.

1 2 3 4 5

I have difficulty maintaining an appropriate speaking pace.

1 2 3 4 5

I sometimes use filler words (e.g., “um”, “uh,” or “like”) when speaking.

1 2 3 4 5

I find it difficult to identify weaknesses in my own speech.

1 2 3 4 5

I would benefit from receiving feedback after practicing a speech.

1 2 3 4 5

Section 3 — Desired Features

Which features would be most useful to you? (Select all that apply)

- ☐ Automatic speech transcription
- ☐ Filler-word detection
- ☐ Pitch and vocal variation analysis
  (Add an explanation)
- ☐ Personalized feedback
- ☐ Speech practice history
- ☐ Suggestions for improvement

How likely are you to use a system that analyzes your speech and provides feedback?

- Very likely
- Likely
- Unsure
- Unlikely
- Very unlikely

Section 4 — Open-Ended Questions

1. What is your biggest challenge when delivering a speech or presentation?
2. What kind of feedback would help you improve your speaking skills?

GOOGLE FORMS 2: ALON SYSTEM EVALUATION

Purpose: Evaluate the actual system after participants have used it.

[DRAFT] This questionnaire gathers feedback about the usability and usefulness of Alon after participants have tested the system. Please answer based on your experience using the application.

Why this target audience?
Why these speech features?
Where did the reading materials come from?
What makes those passages valid for assessment?
How will Alon know that something is actually a problem?
How will you test whether the detection is correct?
What existing system or literature supports the design?
How does the coaching flow actually help the user improve?